LESSON PLAN

BASIC ELECTRICAL (23)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	12
Total No. of Lesson Plans	29	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Shock observers, types, brief descriptions, operation of oleos
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Shock observers, types, brief descriptions, operation of oleos
5. Specific Objectives	23.1.3.1 Trainee will be able to describe typical alighting gears of an aircraft and its construction.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of oleo strut cutaway diagrams and identification of shock observer components from L & Ds; Instructor supervision required for diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the landing gear shock absorber?
(b) Name two common types of shock observers used in aircraft.
(c) How does an oleo strut differ from a simple spring strut?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Shock observer function and necessity • Types of shock observers • Brief description of oleo strut • Operation of oleo strut compression • Operation of oleo strut extension	L & Ds	PPT, IB	15 min
(b)	• Detailed construction of oleo strut • Oil flow during compression stroke • Oil flow during extension stroke • Role of check valve in oleo • Role of snubber in oleo	L & Ds	PPT, PDF	20 min
(c)	• Comparison of shock observer types • Maintenance considerations for oleos • Common failure modes in shock observers • Impact of shock observer failure • Summary of oleo operation principles	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the sequence of oil flow during the compression phase of an oleo strut.
(b) Describe the specific role of the snubber in the extension phase of an oleo strut.
(c) Differentiate between the operation of a shock observer and a simple spring strut.

12. Home Assignment and Follow Up	Prepare a brief note on the design requirements for the next lesson (LP 13) regarding brake units, focusing on how they interface with the landing gear shock system.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Types of brake unit, general design requirements and brief description of a typical brake unit, friction discs and pads, brake master cylinders	Refer to the master syllabus reference list	LP 13

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min